

VERITASFi: An Adaptable, Multi-tiered RAG Framework for Multi-modal Financial Question Answering

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Abstract

Retrieval-Augmented Generation (RAG) is becoming increasingly essential for Question Answering (QA) in the financial sector, where accurate and contextually grounded insights from complex public disclosures are crucial. However, existing financial RAG systems face two significant challenges: (1) they struggle to process heterogeneous data formats, such as text, tables, and figures; and (2) they encounter difficulties in balancing general-domain applicability with company-specific adaptation. To overcome these challenges, we present **VERITASFi**, an innovative hybrid RAG framework that incorporates a multi-modal preprocessing pipeline alongside a cutting-edge two-stage training strategy for its re-ranking component. VERITASFi enhances financial QA through three key innovations: (1) **A multi-modal preprocessing pipeline** that seamlessly transforms heterogeneous data into a coherent, machine-readable format. (2) **A tripartite hybrid retrieval engine** that operates in parallel, combining deep multi-path retrieval over a semantically indexed document corpus, real-time data acquisition through tool utilization, and an expert-curated memory bank for high-frequency questions, ensuring comprehensive scope, accuracy, and efficiency. (3) **A two-stage training strategy** for the document re-ranker, which initially constructs a general, domain-specific model using anonymized data, followed by rapid fine-tuning on company-specific data for targeted applications. By integrating our proposed designs, VERITASFi presents a **novel framework** that greatly enhances the adaptability and robustness of financial RAG systems, providing a scalable solution for both general-domain and company-specific QA tasks. Code accompanying this work is available at <https://github.com/simplew4y/VeritasFi.git>.

CCS Concepts

• Information systems → Question answering.

Keywords

LLM, Information Retrieval, Document Pre-Processing, Large Language Models, Retrieval Augmented Generation, Generative Models, Synthetic Datasets, Generative Retrieval

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1 Introduction

The integration of Large Language Models (LLMs) into the financial sector has catalyzed the development of sophisticated Question Answering (QA) systems, with Retrieval-Augmented Generation (RAG) rapidly emerging as a cornerstone technology [15, 35].

By augmenting LLMs with external knowledge from complex corporate disclosures, such as U.S. Securities and Exchange Commission (SEC) filings, RAG frameworks aim to deliver accurate, verifiable, and context-aware insights essential for high-stakes decision-making and regulatory compliance [8, 12, 19, 32, 49]. For example, when financial analysts query a company's 10-K report for "risks associated with supply chain disruptions in the last fiscal year", they need more than a simple keyword match. They require a synthesized response that integrates complex textual descriptions, numerical data from tables, and trends depicted in figures.

However, deploying general-purpose RAG in finance remains difficult. Two gaps persist: (i) handling the *multi-modal* nature of filings (text/tables/figures), and (ii) balancing general-domain knowledge with entity-specific adaptation. Many RAG pipelines and financial extractors linearize documents as flat text, leading to incomplete grounding [5, 15, 17, 38]. Domain shift is well documented: zero-shot general rerankers degrade across datasets [46], while in-domain query generation and finetuning improve reranking [2, 11]; vertical models further reinforce this (biomedical [18, 34], legal [3, 4], finance [1]).

To overcome these fundamental limitations, we introduce VERITASFi, an adaptable, hybrid RAG framework designed for robust and precise financial question answering. At its core, VERITASFi combines a multi-modal preprocessing pipeline with a Tripartite Hybrid Retrieval (THR) engine and a two-stage Domain-to-Entity re-ranking strategy. In extensive experiments, VERITASFi attains the best end-to-end performance across diverse financial QA datasets, outperforming strong baselines including GraphRAG and LighTRAG, while remaining superior even when classical retrievers (BM25/Faiss) are strengthened with CACK preprocessing. Beyond end-to-end accuracy, our multi-path retrieval delivers higher evidence hit rates, while the two-stage re-ranking improves retrieval quality and enables rapid, low-overhead adaptation to new companies.

In summary, our key contributions are as follows:

- We propose **VERITASFi**, a novel, end-to-end RAG framework that integrates multi-modal data processing, hybrid retrieval, and an adaptable re-ranker to address the challenges of financial QA.
- We introduce a **multi-modal preprocessing pipeline** that effectively unifies disparate data formats from financial filings, which include text, tables, and figures, into a coherent, machine-readable format optimized for retrieval.
- We design a **tripartite hybrid retrieval engine** that combines deep multi-path retrieval, a high-frequency question-matching cache, and external tool integration to ensure comprehensive scope, low latency, and factual timeliness.
- We develop a **two-staged, domain-to-entity adaptation strategy** that first builds a generalized financial re-ranker and then rapidly specializes it for a target entity using automated data annotation, which promises a scalable and rapid solution to model customization for onboarding.

- The extensive experiments on both public benchmarks and in-house company dataset showcase that VERITASFi significantly outperforms existing RAG architectures in correctness, context relevance, and overall answer quality.

2 System Overview

Tackling the complexities of domain-specific financial question answering requires a system that can synthesize knowledge from highly heterogeneous sources, including multi-modal documents, structured time-series data, and real-time external information. To this end, we propose an advanced hybrid RAG framework designed to meet these demands. The core components of our system, which operate synergistically, are: **i) a Context-Aware Knowledge Curation (CAKC) module** to process unstructured financial filings; **ii) a Tripartite Hybrid Retrieval (THR) engine** that queries diverse information sources in parallel.

3 Context-Aware Knowledge Curation

The CAKC module converts raw, multi-modal English financial documents into structured knowledge bases optimized for efficient retrieval. As shown in Figure 1, its primary goal is to decompose complex filings into discrete, contextually coherent information units for indexing. The pipeline yields two complementary outputs.

- (1) **SEC Filing Corpus:** A primary corpus consisting of semantically enriched, self-contained English *text chunks*, transformed and enhanced from SEC filings.
- (2) **High-frequency memory bank:** A structured, factual cache containing *time-stamped answers* to quantitative questions, derived from a semi-automated, human-in-the-loop process.

3.1 Textual Transformation of Multi-Modal Documents

Our pipeline first parses each multi-modal SEC filing document into a sequence of fixed-length chunks (200 words) with different modality (i.e., text, figure, or table), via the open-source tool MinerU [21, 47, 48, 55]. To create a unified textual representation suitable for LLM, we then employ GPT-4o [42] to transform all non-textual content into descriptive narratives. Specifically, figures are converted into structured captions summarizing their key insights, while tables are rewritten as textual statements that capture their primary data trends and relationships.

3.1.1 Semantic Enhancements. To ensure that the resulting textual chunks are coherent and optimized for retrieval, we perform three semantic enhancements:

- (1) **Similar chunk de-duplication.** To reduce content redundancy, we conduct pair-wise comparison that discard any chunk whose dense embedding similarity with other chunks exceeds a predefined threshold τ_{sim} . Specifically, we map each text chunk c_i to the dense representation of the embedding model $e_i = E_{bge_m3}(c_i)$, and measure pairwise semantic relationship between text chunks with cosine similarity $\text{CosSim}(\cdot, \cdot)$. The final de-duplicated corpus C' is then constructed by iteratively removing the subsequent similar chunk c_j :

$$C' = C \setminus \{c_j \in C \mid \forall i \neq j : \text{CosSim}(e_i, e_j) > \tau_{sim}\}, \quad (1)$$

- (2) **Co-reference resolution.** To enhance the contextual sufficiency of each chunk, we replace the pronouns in a given chunk

with their explicit antecedents, as chunks retrieved in isolation lack the necessary context to resolve pronouncements, where such ambiguity leads to factual inaccuracies in generation.

- (3) **Metadata generation.** To ensure individual chunks retain the broader context from other chunks in the same parent section for retrieval, we generate a section-level summary with GPT-4o as a metadata attachment to every chunk. This creates a shared contextual anchor for high-level queries.

3.2 High Frequency Memory Bank Generation

Beyond processing unstructured financial filings, the CAKC pipeline pre-caches a High-Frequency Memory Bank to accelerate common quantitative query lookups. We structure the cache as a table where each row corresponds to different canonical, normalized questions curated by financial experts. The columns represent time points (e.g., fiscal quarters), and the cells record the value of question's subject at the specified time point. We initialize the memory bank by running our full RAG pipeline on source filings for each question, where we utilize DeepSeek R1 for questions requiring complex reasoning across multiple retrieved chunks. The initialized memory bank is then carefully verified by in-house experts, resulting in a highly reliable, structured knowledge base. Details on the canonical questions and the table structure can be found in Appendix C.

3.3 Knowledge Base Indexing

We index all processed text chunks in parallel for semantic retrieval. Dense representations e_i are mapped through $E_{bge_m3}(\cdot)$ and stored in Chroma vector database [28]. Sparse representations are built by inverted index over the text chunks. The High-Frequency Memory Bank is indexed in a relational database to allow on-the-fly subsequence and BM25 searches against its canonical questions. The canonical questions are embedded and indexed within the vector datastore to facilitate semantic matching.

4 Tripartite Hybrid Retrieval

To provide comprehensive answers, we first run a *Query Preprocessing* stage that normalizes, disambiguates, and decomposes the user query into minimal self-contained sub-queries. Building on this, our **Tripartite Hybrid Retrieval (THR) engine** (Figure 2) routes the processed request to three specialized modules in parallel to handle financial queries of different complexities and data types: (1) **Multi-Path Retrieval with Domain-to-Entity Adaptation Re-ranking** for deep semantic search over unstructured filings; (2) **High-Frequency Memory Look-up** for rapid answers from a pre-cached memory bank; and (3) **Tool Use** for real-time data via external APIs. Evidence from all three streams is then fused to synthesize the final answer.

4.1 Query Preprocessing

This module serves as the entry point to the retrieval pipeline, aims to transform complex, multi-faceted financial queries, which rely on unspoken context from previous turns, into precise, machine-actionable instructions. The contextual transformation is achieved in two steps:

- (1) **Query Normalization and Context Augmentation** tackles the problem of incompleteness and ambiguity. We normalize

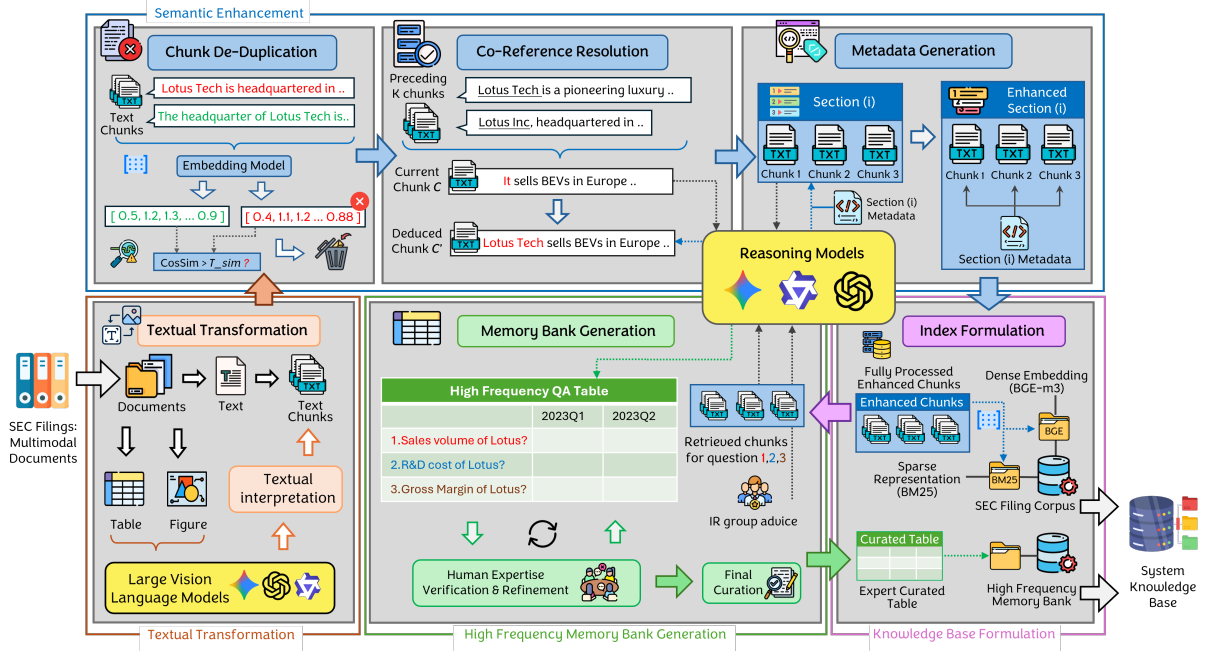


Figure 1: Context-Aware Knowledge Curation (CAKC) pipeline. Textual Transformation Module (Orange) transforms multi-modal data into textual representation. Semantic Enhancement module (Blue) refines the text chunks. The Knowledge Base Formulation Module (Purple) generates the final SEC Filing Corpus from processed text chunks, and the High Frequency Memory Bank Generation Module (Green) constructs a High-Frequency Memory Bank of time-stamped answers.

the raw query by translating queries into English to align its representation with the semantic space of our English-language corpus. Next, this pipeline includes the entire conversation history for co-reference resolution and context augmentation, allowing LLM to integrate prior turns and identify users' complete intent, especially when current turn is highly elliptical or dependent on previous information.

- (2) **Intent-Driven Decomposition and Retrieval Router** tackles the problem of data complexity and execution efficiency. We decompose enhanced, context-rich query into smallest possible self-contained query units (sub-queries), and assign each to optimal retrieval paths using the High-Frequency Memory Bank, pre-defined Tools, deep retrieval from document corpus, or direct answering using LLM's own knowledge.

4.2 Multi-Path Retrieval

For sub-queries that require insights from financial filings, the system activates the Multi-Path Retrieval and Domain-to-Entity Adaptation Re-ranking stream. This pipeline is designed to maximize information recall by using diverse retrieval strategies to gather a broad set of candidate chunks.

4.2.1 Retrieval Module. The Retrieval Module executes three distinct strategies in parallel to retrieve from the corpus, each targeting a different aspect of the query-chunk relationship: (1) **BM25 Sparse Retriever:** A lexical search model that calculates chunk keyword overlap with the query using the Best-Matching 25 (BM25) scoring function. (2) **Dense Retriever:** A semantic retriever that encodes

queries and document chunks into dense vectors. The relevant documents are retrieved via cosine similarity, using FAISS [29] nearest-neighbor search. (3) **Metadata Retriever:** A high-level semantic search that matches the query with LLM-generated summaries attached as metadata to each chunk during the CAKC stage.

4.2.2 Chunk Bundling Module. To capture complete semantic context, this module expands each retrieved chunk c_i by iteratively appending neighbors adjacent in the original passage order based on semantic similarity. For each retrieved chunk, the module starts from an initial bundle $B_i = \{c_i\}$, it examines up to k chunks in both directions. A candidate chunk $c_j \in \{c_{i-k}, \dots, c_{i-1}, c_{i+1}, \dots, c_{i+k}\}$ is added to the bundle if

$$\text{CosSim}(e_i, e_j) > \tau_{\text{bundle}} \quad \text{and} \quad c_j \notin B_i, \quad (2)$$

where $e_i = E_{bge_m3}(c_i)$, and $\text{CosSim}(\cdot, \cdot)$ denotes the cosine similarity function, resulting in the expanded bundle $B_i = B_i \cup \{c_j\}$. This expansion ensures that multi-paragraph discussions in financial documents are captured cohesively.

4.3 Domain-to-Entity Adaptation Re-ranking

To refine candidates from the previous stages, the Domain-to-Entity Adaptation Re-ranking (DAR) module filters and re-orders chunks to improve precision. A decoder-only model scores each chunk's relevance to the query using a prompt-based binary classification approach [36]. Given a query q and candidate chunk c , the model classifies the chunk as relevant ("yes") or not ("no"). In inference, the final relevance score is the probability assigned to the "yes"

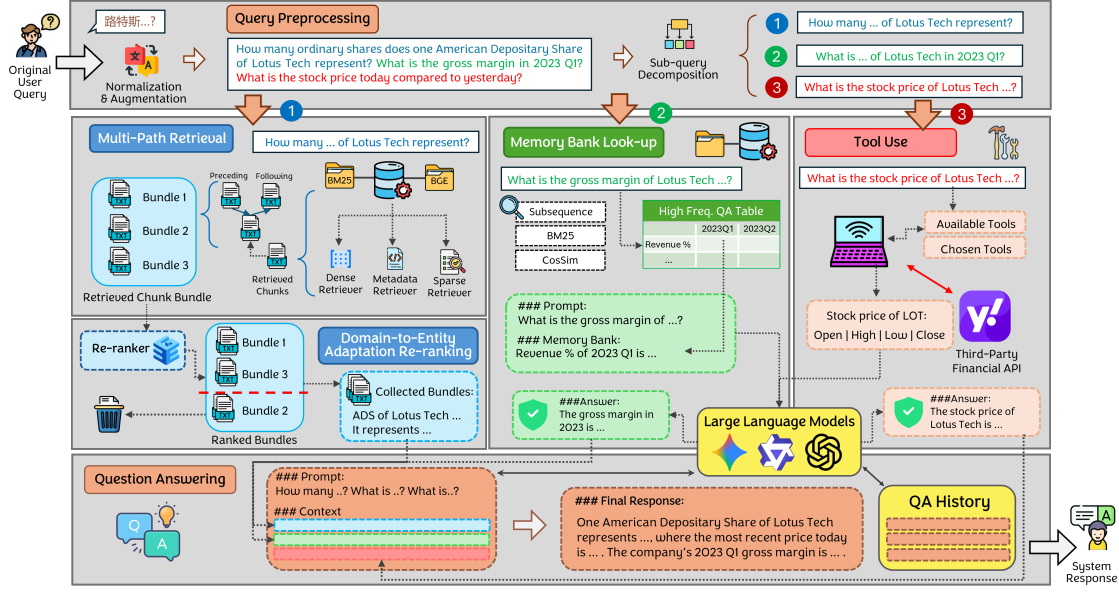


Figure 2: The complete VERITASFi RAG pipeline. The diagram shows three concurrent knowledge retrieval paths: Multi-Path Retrieval (Blue), Memory Bank Look-up (Green) and Tool Use (Red).

class, computed as:

$$\text{Score}(q, c) = \mathbb{P}(\text{relevant} \mid q, c) = \sigma(z_{\text{yes}} - z_{\text{no}}) \quad (3)$$

where z_{yes} and z_{no} are the logits for “yes”/“no,” and $\sigma(\cdot)$ is the sigmoid. During training, a supervised contrastive loss on the “yes”/“no” logits guides the re-ranker to better distinguish relevant chunks across general and entity-specific domains.

4.3.1 Domain-to-Entity Adaptation Strategy. We propose a two-stage, domain-to-entity fine-tuning strategy to develop a re-ranker that is both financially knowledgeable and rapidly adaptable to specific entities. The first stage trains a general financial re-ranker on an abstracted, entity-agnostic dataset, while the second stage specializes the model for a target company using automatically annotated data. This tiered approach maximizes base-model generalization while minimizing manual customization effort.

Stage 1: Finance Reranker Training. The first stage aims to train a re-ranker that captures the semantic and structural nuances of financial filings. Using a high-quality, expert-annotated dataset, we apply data augmentation and systematically mask entity-specific information via three abstraction strategies: i) Replace product models and individual names (e.g., executives, board members) with placeholders. ii) Replace the target company with a placeholder and other companies with randomly sampled competitors. iii) Apply full abstraction, replacing all company, product, and individual mentions with consistent placeholders. Training on augmented corpora enables the re-ranker to perform financial reasoning independent of memorized facts, yielding a general-purpose model. The optimal abstraction strategy is selected based on comparative performance.

Stage 2: Target Company Adaptation. The second stage specializes the general re-ranker for a target company. To eliminate the manual data creation bottleneck, we use an automated annotation

module where an LLM labels retrieved chunks as relevant or irrelevant. This process operates on the actual output of our Multi-path Retrieval pipeline, ensuring the training data distribution mirrors the inference-time conditions. The annotator’s judgment is guided by few-shot examples and defines clear relevance criteria (e.g., direct answer, contextual support, fuzzy time match). This automated approach ensures the consistent and scalable generation of high-quality training data for specializing the re-ranker to any new company. Details on the annotation prompt are in Appendix D.

4.3.2 Contrastive Learning Objective. Both stages use supervised fine-tuning (SFT) with a contrastive learning objective. For a query q , the model is trained to assign higher scores to positive (relevant) chunks p^+ than to K negative (irrelevant) chunks $\{p_j^-\}_{j=1}^K$.

For training, all chunks retrieved for q are annotated: relevant chunks form the positive set $\mathcal{P}_q = \{p_1^+, \dots, p_n^+\}$, and irrelevant chunks form the negative set $\mathcal{N}_q = \{p_1^-, \dots, p_m^-\}$. To enhance robustness, $\mathcal{N}_q$ combines (1) hard negatives from retriever-identified but irrelevant chunks, and (2) random negatives sampled from the corpus [25]. Each training quadruple is structured as

$$\mathcal{D} = \{ \langle q, \mathcal{P}_q, \mathcal{N}_q, \text{prompt} \rangle \} \quad (4)$$

For each query, a mini-batch contains one positive $p^+ \in \mathcal{P}_q$ and K negatives $\{p_1^-, \dots, p_K^-\} \subset \mathcal{N}_q$, producing scores $s = [s^+, s_1^-, \dots, s_K^-]$ with $s_i = \text{Score}(q, p_i)$.

The $(K + 1)$ candidates are treated as an *ad-hoc* set of classes, converting the task into classification where p^+ is the positive class. The training loss is standard cross-entropy:

$$\mathcal{L} = -\log \frac{e^{s^+}}{e^{s^+} + \sum_{j=1}^K e^{s_j^-}} \quad (5)$$

For Stage 1, negatives in the human-annotated dataset $\mathcal{D}_{\text{human}}$ serve as hard negatives. Stage 2 uses hard negatives generated by

Procedure 1: Domain-to-Entity Re-ranker Fine-Tuning**Input:** Human-annotated general dataset

$$\mathcal{D}_{\text{human}} = \{(q, p^+, p^-), \dots\}$$

where q is a query, p^+ is a relevant chunk (positive), and p^- is an irrelevant chunk (negative);

Target company queries $\mathcal{Q}_{\text{target}}$ and corpus $\mathcal{C}_{\text{target}}$;

Base re-ranker model $\mathcal{M}_{\text{base}}$, Retriever $\mathcal{R}$, LLM-Annotator $\mathcal{L}_{\text{anno}}$;

Output: Specialized re-ranker model $\mathcal{M}_{\text{specialized}}$

```

1 Function TrainContrastive( $\mathcal{M}, \mathcal{D}_{\text{train}}$ ):
2   foreach training step do
3     Sample a query  $q$  from  $\mathcal{D}_{\text{train}}$ ;
4     Let  $\mathcal{P}_q, \mathcal{N}_q$  be the positive and negative sets for  $q$ ;
5     Sample one positive chunk  $p^+ \in \mathcal{P}_q$ ;
6     Sample  $K$  negative chunks  $\mathcal{N}^- \subset \mathcal{N}_q$ ;
7     Compute logits  $\mathbf{s} = [\text{Score}(q, p^+), \dots, \text{Score}(q, p_K^-)]$ 
        with model  $\mathcal{M}$ ;
8     Compute loss  $\mathcal{L} = -\log \left( \frac{e^{s^+}}{e^{s^+} + \sum_{j=1}^K e^{s_j^-}} \right)$ ;
9     Update  $\mathcal{M}$  by minimizing  $\mathcal{L}$ ;
10  return  $\mathcal{M}$ 

/* Stage 1: Financial Re-ranker Training. */
11 ;
12  $\mathcal{D}_{\text{aug}} \leftarrow \emptyset$ ;
13 foreach  $(q, p^+, p^-) \in \mathcal{D}_{\text{human}}$  do
14    $q_{\text{aug}}, p_{\text{aug}}^+, p_{\text{aug}}^- \leftarrow \text{ApplyEntityAbstraction}(q, p^+, p^-)$ ;
15   Add  $(q_{\text{aug}}, p_{\text{aug}}^+, p_{\text{aug}}^-)$  to  $\mathcal{D}_{\text{aug}}$ ;
16  $\mathcal{M}_{\text{general}} \leftarrow \text{TrainContrastive}(\mathcal{M}_{\text{base}}, \mathcal{D}_{\text{aug}})$ ;

/* Stage 2: Target Company Adaptation */
17 ;
18  $\mathcal{D}_{\text{auto}} \leftarrow \emptyset$ ;
19 foreach  $q \in \mathcal{Q}_{\text{target}}$  do
20    $C_q \leftarrow \mathcal{R}(q, \mathcal{C}_{\text{target}})$ ;
21    $\mathcal{P}_q, \mathcal{N}_{\text{hard}} \leftarrow \mathcal{L}_{\text{anno}}(q, C_q)$ ; // hard negatives
22    $\mathcal{N}_{\text{rand}} \leftarrow \text{SampleRandomChunks}(\mathcal{C}_{\text{target}})$ ; // random
        negatives
23    $\mathcal{N}_q \leftarrow \mathcal{N}_{\text{hard}} \cup \mathcal{N}_{\text{rand}}$ ;
24   Add  $(q, \mathcal{P}_q, \mathcal{N}_q)$  to  $\mathcal{D}_{\text{auto}}$ ;
25  $\mathcal{M}_{\text{specialized}} \leftarrow \text{TrainContrastive}(\mathcal{M}_{\text{general}}, \mathcal{D}_{\text{auto}})$ ;
26 return  $\mathcal{M}_{\text{specialized}}$ 

```

LLM annotator. Training is implemented with FlagEmbedding¹ to fine-tune bge-reranker-v2-gemma [6]. Full procedure and hyper-parameters are in Procedure 1 and Appendix B.

4.4 Memory Bank Look-up

The Memory Bank Look-up module provides instantaneous answer to common financial queries with the pre-trained Memory Bank

¹<https://github.com/FlagOpen/FlagEmbedding>

(Sec. 3.2). After matching the incoming user query with the pre-compiled knowledge base, the system can bypass computationally intensive deep retrieval, reasoning and generation steps, hence reducing overall latency. We normalize the query to remove punctuation and irrelevant entities, and match the query to knowledge base with the three parallel algorithms: (1) **Subsequence-based Similarity** computes structural similarity between the query and canonical questions. It identifies the longest contiguous matching subsequences of characters between two strings to capture precise terminology and word order (sub-sequence). (2) **Keyword-based Similarity (BM25)** handles variations in phrasing by applying the BM25 algorithm on structure table. We measure the relevancy of a canonical question using the scoring function discussed in Section 4.2. (3) **Semantic Similarity** tackles queries that are syntactically different but semantically equivalent to the canonical questions, aligning them via the dense retrieval in Section 4.2.

We consider a canonical question c_k potential match if its score from any of the three methods $S_{\text{seq}}(q, c_k)$, $S_{\text{bm25}}(q, c_k)$, $S_{\text{sem}}(q, c_k)$ surpasses the confidence thresholds τ_{seq} , τ_{bm25} , and τ_{sem} . The matching for (q, c_k) is determined by the boolean function $M(q, c_k)$:

$$M(q, c_k) = (S_{\text{seq}}(q, c_k) > \tau_{\text{seq}}) \vee (S_{\text{bm25}}(q, c_k) > \tau_{\text{bm25}}) \vee (S_{\text{sem}}(q, c_k) > \tau_{\text{sem}}) \quad (6)$$

If $M(q, c_k)$ is true for any c_k , the system retrieves canonical question c_k and the corresponding entry from memory bank to return the answer in a structured key-value format.

4.5 Tool Use

While retrieval from SEC filings covers historical data, queries requiring real-time information, such as current stock prices or recent corporate actions, need external sources. The Tool Use module addresses this gap by integrating a third-party financial API to fetch live market data and access event-specific information (e.g., IPO details). The module operates asynchronously alongside other retrieval streams and adheres to OpenAI's function-calling schema to autonomously invoke appropriate tools based on query requirements. Upon execution, tool outputs are integrated into the conversation as additional evidence.

5 Experiments

5.1 Datasets and Baseline

We evaluate VERITASFi across 4 financial QA datasets: **FinanceBench** [27], containing human-verified QA pairs with single gold reference chunks from financial filings; **FinQA** [8], from which we use the textual reasoning subset with single reference chunks; and 2 **In-House Company Dataset** with multi-entity (e.g., Lotus, Zeekr) QA pairs and annotated relevance labels. We compare our model against graph-based RAG systems **GraphRAG** and **LightRAG** [20], as well as standard direct retrieval-to-generation baselines.

5.2 Experiment Settings

5.2.1 Multi-path Retrieval (MPR) settings. We perform retrieval tasks on both in-house company dataset and open-source dataset Financebench [27] for MPR module. Preprocessing consists of: (1) segmenting filings into chunks using CAKCC, (2) indexing the chunks in a vector store, and (3) optimize user queries for RAG. To ensure a

Table 1: End-to-End Performance (top-10). Four panels (2×2). CAKC = Context-Aware Knowledge Curation; Correct. = Correctness; Resp. Rel. = Response Relevancy; Ctx. Recall/Prec. = Context Recall/Precision. Avg. is the unweighted mean of {Correct., Resp. Rel., Ctx. Recall, Ctx. Prec.}. Values are %. Our method is in bold.

Zeekr						
w/ CAKC	Config	Correct.	Resp. Rel.	Ctx. Recall	Ctx. Prec.	Avg.
No	GraphRAG	23.3	38.9	0.1	11.7	18.5
No	LightRAG	20.5	46.8	11.2	19.3	24.5
No	BM25	6.0	16.2	3.8	27.2	13.3
No	Faiss	3.8	12.0	2.3	15.7	8.4
No	BM25+Faiss	5.6	15.0	3.1	23.2	11.7
Yes	BM25	29.4	63.0	30.3	44.7	41.9
Yes	Faiss	37.4	69.6	36.5	41.0	46.1
Yes	BM25+Faiss	35.9	69.5	36.0	41.1	45.6
Yes	VERITASFi (ours)	64.3	82.0	59.0	46.3	62.9

FinanceBench						
w/ CAKC	Config	Correct.	Resp. Rel.	Ctx. Recall	Ctx. Prec.	Avg.
No	GraphRAG	9.2	6.8	2.6	2.7	5.33
No	LightRAG	40.3	6.5	3.1	8.8	14.7
No	BM25	17.7	54.3	34.6	39.8	36.6
No	Faiss	13.7	51.3	30.4	40.6	34.0
No	BM25+Faiss	18.6	53.4	32.5	46.8	37.8
Yes	BM25	22.8	32.6	29.4	15.3	25.0
Yes	Faiss	20.9	63.2	50.8	35.9	42.7
Yes	BM25+Faiss	21.7	52.8	46.0	28.1	37.1
Yes	VERITASFi (ours)	24.7	75.1	49.2	37.2	46.5

Lotus						
w/ CAKC	Config	Correct.	Resp. Rel.	Ctx. Recall	Ctx. Prec.	Avg.
No	GraphRAG	25.9	43.8	0.4	8.2	19.6
No	LightRAG	24.8	58.8	15.9	28.7	32.1
No	BM25	17.5	34.2	7.0	23.5	20.6
No	Faiss	16.0	31.3	6.5	20.6	18.6
No	BM25+Faiss	16.0	22.4	6.4	23.7	17.1
Yes	BM25	39.6	79.0	37.2	44.0	50.0
Yes	Faiss	39.7	81.4	45.5	40.4	51.8
Yes	BM25+Faiss	46.5	77.9	48.2	46.8	54.9
Yes	VERITASFi (ours)	57.3	89.4	57.6	51.6	64.0

FinQA Bench						
w/ CAKC	Config	Correct.	Resp. Rel.	Ctx. Recall	Ctx. Prec.	Avg.
No	GraphRAG	22.8	53.1	4.6	2.1	25.8
No	LightRAG	29.5	64.1	34.0	33.4	40.3
No	BM25	30.3	54.0	32.4	40.8	39.4
No	Faiss	28.2	51.1	29.1	39.6	37.0
No	BM25+Faiss	30.9	53.4	32.1	45.2	40.4
Yes	BM25	31.4	72.6	73.3	27.8	51.3
Yes	Faiss	33.4	76.8	75.7	29.1	53.7
Yes	BM25+Faiss	32.0	74.8	74.7	29.6	52.8
Yes	VERITASFi (ours)	33.7	80.1	73.9	38.8	56.6

fair comparison, we maintain a comparable total number of unique retrieved chunks for each setting. Retrieval quality is measured by the evidence hit rate, where a retrieved chunk is classified as a hit if its cosine similarity to the ground-truth evidence exceeds the threshold. We use Qwen/Qwen-Embedding-4B [45] as our embedding model. For in-house dataset, this threshold is set to 1.0, as the ground-truth evidence chunks are the exact chunks selected from the vector store.

5.2.2 Domain-to-Entity Adaptation Re-ranking (DAR) settings. The re-ranker evaluation is designed to rigorously test two distinct performance objectives: general financial domain competence and target company specialization. The performance of the model is evaluated using metrics at K=5 and K=10: NDCG@K, MRR@K, Precision@K, and Recall@K. The evaluation is structured around a two-stage benchmarking process, with specific baselines defined for each stage to quantify the performance gain. **Stage 1** assesses the model’s ability to generalize across financial entities. The re-ranker is trained on the union of abstracted dataset from both Lotus and Zeekr. The performance baseline for Stage 1 is the re-ranker’s own pretrained version. We evaluate the three abstraction strategies and select the checkpoint with the highest Precision@5 as the optimal general model as well as the base model for Stage 2 fine-tuning. **Stage 2** assesses the specialization for the target company, in which the baseline is defined by a control experiment: the re-ranker’s pretrained version directly fine-tuned with LoRA [23] on the company-specific dataset (the checkpoint that achieved the highest Precision@5 in this control run). This allows us to quantify the superiority of the two-stage approach over immediate specialization.

5.2.3 End-to-End response settings. To provide a holistic end-to-end assessment of generation quality of VERITASFi, we adopt four metrics from the RAGAS [14]: Factual Correctness, which measures the factual alignment of the generated response with the answer; Response Relevancy, which assesses how directly the response addresses the user query; Context Recall, which quantifies the coverage of relevant information in the retrieved context; and Context Precision, which evaluates the proportion of relevant information among the retrieved passages. Metric calculation details are provided in Appendix F, while configurations for other baselines (e.g., GraphRAG, LightRAG) are detailed in Appendix G.

6 Results and Analysis

6.1 End-to-end Response Results

Results in Table 1 demonstrate VERITASFi’s leading performance across all datasets. The CAKC module substantially enhances all methods, particularly in Context Recall and Factual Correctness. On proprietary datasets (Zeekr, Lotus), VERITASFi achieves superior accuracy and intent alignment. Collectively, these findings confirm the robustness and generalizability of our integrated QA architecture.

6.2 Multi-path Retrieval Results

Our multi-path approach consistently leads across all datasets (Figure 3). By combining four methods, the ensemble achieves a 2-5% hit rate gain over the competitive Faiss-only baseline. Unlike individual hybrid methods (e.g., Faiss+BM25) which fluctuate across

Table 2: Reranker performance on company dataset under different retrieval settings (SFT with Zeekr + Lotus Data). Green gains next to Stage 1 indicate improvements over the pretrained baseline. Dark-green gains next to S2 (Stage 2) indicate further improvements over the best S1 (Stage 1: Complete Abstraction) variant.

Dataset	Base	Training Data	NDCG@5	MRR@5	Precision@5	Recall@5	NDCG@10	MRR@10	Precision@10	Recall@10
Lotus	PT	N/A (Baseline)	62.0	60.1	29.3	23.0	67.3	68.2	24.4	34.8
	PT+S1	w/ Product/Person Abs. D.	64.2 (↑ 2.2)	62.7 (↑ 2.6)	32.2 (↑ 2.9)	25.8 (↑ 2.8)	69.3 (↑ 2.0)	69.0 (↑ 0.8)	26.7 (↑ 2.3)	38.5 (↑ 3.7)
		w/ Company Name Abs. D.	65.1 (↑ 3.1)	65.2 (↑ 5.1)	31.1 (↑ 1.8)	24.2 (↑ 1.2)	69.5 (↑ 2.2)	71.5 (↑ 3.3)	25.6 (↑ 1.2)	37.3 (↑ 2.5)
		w/ Complete Abs. Dataset	69.2 (↑ 7.2)	69.1 (↑ 9.0)	33.4 (↑ 4.1)	26.1 (↑ 3.1)	69.2 (↑ 1.9)	71.0 (↑ 2.8)	27.1 (↑ 2.7)	38.4 (↑ 3.6)
	PT+S2	Company Specific Dataset	72.6 (↑ 3.4)	71.5 (↑ 2.4)	35.3 (↑ 1.9)	27.9 (↑ 0.4)	71.5 (↑ 2.3)	73.4 (↑ 2.4)	27.5 (↑ 0.4)	39.0 (↑ 0.6)
Zeekr	PT+S1+S2	Company Specific Dataset	75.1 (↑ 5.9)	73.5 (↑ 4.4)	37.4 (↑ 4.0)	28.3 (↑ 2.2)	75.4 (↑ 6.2)	78.6 (↑ 7.6)	28.1 (↑ 1.0)	39.5 (↑ 1.1)
	PT	N/A (Baseline)	77.1	70.0	64.9	32.1	74.2	65.3	57.4	43.1
	PT+S1	w/ Product/Person Abs. D.	79.8 (↑ 2.7)	73.4 (↑ 3.4)	67.9 (↑ 3.0)	34.2 (↑ 2.1)	74.2 (=)	66.3 (↑ 1.3)	59.3 (↑ 1.9)	45.7 (↑ 2.6)
		w/ Company Name Abs. D.	81.6 (↑ 4.5)	75.1 (↑ 5.1)	69.9 (↑ 5.0)	35.0 (↑ 2.9)	75.7 (↑ 1.5)	67.4 (↑ 2.1)	59.5 (↑ 2.1)	44.6 (↑ 1.5)
		w/ Complete Abs. Dataset	82.6 (↑ 5.5)	75.9 (↑ 5.9)	71.5 (↑ 6.6)	35.7 (↑ 3.6)	76.3 (↑ 2.1)	69.4 (↑ 4.1)	59.9 (↑ 2.5)	44.4 (↑ 1.3)
	PT+S2	Company Specific Dataset	84.4 (↑ 1.8)	77.3 (↑ 1.4)	75.5 (↑ 4.0)	38.8 (↑ 3.1)	76.5 (↑ 0.2)	68.4 (↓ 1.0)	61.5 (↑ 1.6)	46.5 (↑ 2.1)
	PT+S1+S2	Company Specific Dataset	87.1 (↑ 4.5)	83.6 (↑ 7.7)	77.2 (↑ 5.7)	39.0 (↑ 3.3)	78.2 (↑ 1.9)	70.9 (↑ 1.5)	63.4 (↑ 3.5)	47.1 (↑ 2.7)

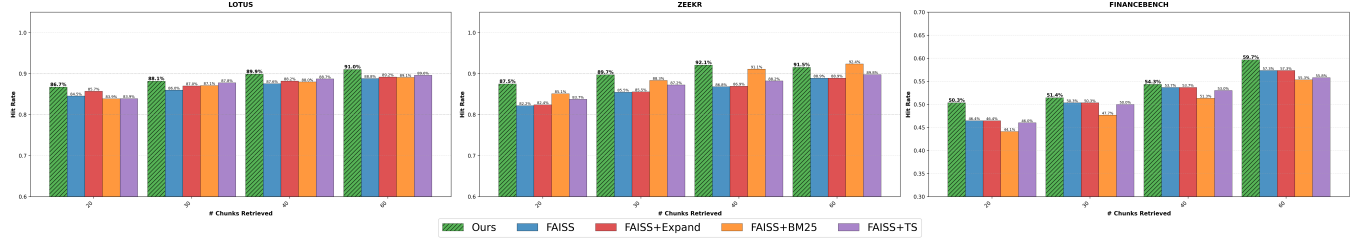


Figure 3: Hit rate comparison across multi-path retrieval (MPR) settings. Our hybrid retrieval approach consistently improves evidence hit rates across evaluation datasets in most retrieval scenarios.

Zeekr and Financebench, our ensemble provides robust and consistent performance by leveraging the complementary strengths of diverse retrieval strategies.

6.3 Domain-to-entity Re-ranking Results

Our two-stage strategy (Section 4.3.1) significantly outperforms baselines across Lotus and Zeekr (Table 2). Stage 1 uses entity-masking to learn robust financial semantics, while Stage 2 refines company-specific relevance. This approach surpasses Direct SFT by >2.5 NDCG@5 and >2.0 MRR@5 on Zeekr, with comparable gains on Lotus. Ultimately, our method demonstrates that initial domain abstraction followed by targeted adaptation yields superior, transferable re-ranking performance compared to standard fine-tuning.

6.4 System Latency and Cost Results

We conduct a comprehensive end-to-end analysis of the system’s operational efficiency, measuring both **system latency** and **deployment cost**. This evaluation assesses the performance of every module in our processing pipeline, specifically quantifying the average execution time and the token usage for each large language model (LLM) invocation. Our analysis particularly focuses on the variable steps, such as **Sub-query Answering** and **Tool Use**, to understand how system performance scales with increasing query

complexity and tool integration. For a detailed breakdown of the time and cost associated with each processing step, please refer to Appendix E.

7 Conclusion

This paper introduced VERITASFi, a practical end-to-end framework for financial QA that addresses the dual challenge of processing heterogeneous disclosures and achieving efficient company-specific adaption. Through the synergy of the *Context-Aware Knowledge Curation*, *Tripartite Hybrid Retrieval* engine and *two-stage Domain-to-Entity re-ranking* training strategy, VERITASFi delivers strong end-to-end performance, with the largest improvement reaching +38.4 points in average on Zeekr compared to LightRAG [20]. Our ablation studies confirm that the three components provide complementary strengths, forming a cohesive and scalable architecture for grounded financial QA. Beyond these empirical gains, VERITASFi establishes a generalizable architectural blueprint for domain-specific RAG systems that must balance broad applicability with rapid specialization—a requirement increasingly critical across sectors beyond finance. Future work could explore extending this framework to other specialized domains (e.g., legal, healthcare) and incorporating advanced reasoning capabilities to handle complex multi-hop queries over temporal financial data.

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A Related Work

A.1 Retrieval Augmented Generation

Large Language Models have demonstrated remarkable capabilities in natural language understanding and generation; however, they face significant challenges in knowledge-intensive tasks, primarily due to their knowledge cutoff and a propensity for hallucination [26]. To address this, the initial RAG framework [35], built upon the Dense Passage Retrieval (DPR) model [30], was introduced to augment LLM responses with relevant document chunks from an external knowledge database. While earlier models like DPR focused on extractive tasks, RAG’s key innovation was extending this retrieval mechanism to generative tasks, enabling the model to produce coherent and factually grounded responses.

Modern RAG systems have evolved from simple retriever-generator architectures into sophisticated multistage pipelines comprising pre-retrieval and post-retrieval processes [15]. The pre-retrieval stage focuses on improving query formulation through techniques such as query rewriting [40] and decomposition, while the post-retrieval stage commonly employs re-ranking to refine the candidate set. Traditional cross-encoder re-rankers [41] jointly encode query-document pairs for fine-grained scoring, whereas LLM-based re-rankers [53] leverage generative reasoning to handle more complex ranking scenarios.

Despite recent advances, RAG systems that rely solely on unstructured text remain limited in capturing complex, multi-hop relationships between entities. To address this, GraphRAG and LightRAG frameworks [13, 20, 22, 24, 39, 44] integrate structured knowledge graphs into retrieval to model inter-entity dependencies and improve factual grounding. While these methods enhance reasoning and coverage, they require expensive graph construction, indexing, and maintenance, making them time-consuming and difficult to scale in dynamic domains.

A.2 RAG in Finance

These limitations are particularly pronounced in the financial domain, where precision and timeliness are critical for regulatory compliance and investment decision-making [16]. Financial documents contain dense terminology, rapidly evolving information, and heterogeneous formats such as tables and reports, making retrieval especially challenging [10, 54]. To address these challenges, prior work explores three main directions. Some methods focus on internal document information, leveraging structural cues or entity co-occurrence to enhance contextual retrieval [5, 17]. Others incorporate external knowledge, such as financial databases or knowledge graphs, to improve factual grounding and reduce hallucination [7]. More recently, hybrid RAG systems attempt to combine both internal and external signals through complex multistage pipelines [31, 33]. While effective, these approaches often require extensive graph construction, corpus refinement, and fine-tuning, leading to high computational cost and limited scalability across financial entities and domains.

A.3 Financial RAG System Evaluation

The systematic evaluation of financial RAG systems is supported by a growing number of specialized benchmarks, largely centered on question answering from SEC filings. A core set of datasets, including Financebench [27], FinQA [8], FinDER [10], and FinBen [50], provide (question, context, answer) tuples with golden evidence, enabling assessment of both retrieval quality and end-to-end accuracy. To address the crucial need for quantitative analysis, benchmarks such as FinQA [8], ConvFinQA [9], TAT-QA [56], and DocFinQA [43] specifically target numerical reasoning over combined textual and tabular data. Further advancing the rigor of evaluation, recent benchmarks introduce greater complexity. T²RagBench [52] uses re-created, context-independent questions to test true retrieval-based reasoning, while SECQUE [51] raises the difficulty with cross-company questions that require synthesis of information across long contexts.

Manually evaluating the outputs of a RAG system is impractical at scale, making automated evaluation essential. Traditional metrics like ROUGE [37] and EM (Exact Match) are often insufficient as they rely on lexical overlap and fail to capture semantic correctness. Consequently, the field has increasingly adopted powerful LLMs as judges. This approach takes two common forms: RAGAS[14] and FinDER [10] prompt an LLM to evaluate a response based on a criteria tuple (faithfulness, correctness, context relevance). Alternatively, as seen in the SECQUE benchmark [51], an LLM can be used to map a response directly to a holistic quality score.

B Re-ranker Fine-tuning

We employ Low-Rank Adaptation (LoRA) [23] through the **FlagEmbedding** framework to reduce the number of trainable parameters and enhance training efficiency. Specifically, LoRA was enabled with the rank set to 32 and the scaling factor (α) set to 64, targeting the attention projection layers (q_{proj} , k_{proj} , v_{proj} , o_{proj}). The model was fine-tuned for 100 epochs. The learning rate was set to $1e-4$ to ensure stable convergence. To manage memory and maintain a small effective batch size, the per-device batch size was set to 1, and gradient accumulation was enabled with *gradient_accumulation_steps*=2. Mixed precision training was employed using the *bf16* setting to further optimize memory usage. The training utilized a warmup ratio of 0.1 and a weight decay of 0.1. Input sequences were configured with a *train_group_size* of 8.

C High Frequency Memory Bank

The memory bank serves as a specialized, low-latency knowledge cache for fact-intensive queries. The illustrative example shown in Figure 4 shows that the key advantage of this mechanism is its ability to directly retrieve accurate, pre-computed answers for frequently asked questions, resulting in minimal end-to-end latency for high-frequency questions.

```
{
  "question": "What were Lotus Technology's R&D expenses in Q1 2024?",
  "answer": "According to the unaudited financial report of Lotus Technology for the first quarter of 2024, its R&D expenses amounted to 1.84692 billion Chinese Yuan. This figure reflects the company's continuous investment in electrification and technological innovation, aligning with its global brand strategy. \n\n(Note: The data source is the Q1 financial report released on May 29, 2024, and currency units have been converted to Chinese Yuan)"
}
```

Figure 4: An example of question-answering using memory bank.

D Automatic Annotation

D.1 Automatic annotation prompt template

Figure 5 presents the complete prompt template used for automatic labeling for retrieved document chunks. This template is designed to guide a Large Language Model (LLM), designated with the Role of an expert financial document annotation specialist, to perform the task of determining the relevance of a given document chunk to a user query. The prompt provides detailed Relevance Assessment Criteria, clearly defining both High Relevance (including Direct Match, Contextual Support, and Fuzzy Time Period Match) and Low Relevance criteria (Generic Discussion and Incidental Mention). Crucially, the prompt incorporates Few-Shot Annotation Examples

to demonstrate the expected relevance judgment and accompanying analysis. Finally, it specifies a strict two-line output format to ensure structured and reliable data generation.

```
# Role
You are an expert financial document annotation specialist, specializing in providing high-quality relevance annotations for question-answering systems focused on financial regulatory filings (such as SEC filings).

# Task
Given a financial domain user query and a document chunk, determine the relevance of the document chunk to the query.

# Relevance Assessment Criteria

**High Relevance (Relevant):**
1. **Direct Answer Match**: The chunk directly contains specific data, figures, or explicit answers required by the query.
2. **Contextual Support**: The chunk provides essential background information or calculation basis necessary to answer the query, even if it doesn't contain the final answer itself.
3. **Fuzzy Time Period Match**: The chunk discusses the same topic but within a different time frame, still providing useful context.

**Low Relevance (Irrelevant):**
1. **Generic Discussion**: Only provides industry background or company overview without supporting the specific query.
2. **Incidental Mention**: Only mentions query elements incidentally in footnotes, disclaimers, or unrelated paragraphs.

# Annotation Examples

**Example 1: Direct Data Match**
Query: What was Lotus Technology's Q4 2023 revenue?
Chunk: "Lotus Technology (LOT) reported Q4 2023 revenue of $750 million, representing a 45% increase year-over-year, primarily driven by strong sales of the Eletre electric vehicle model."
Analysis: The chunk directly provides the required Q4 2023 revenue data ($750 million), completely matching the query requirements.
Relevance: yes
---

**Example 2: Keywords Present but Content Mismatched**
xxx
---

**Example 3: Time Period Mismatch (still relevant)**
xxx
---

# Begin Annotation
Query: {query}
Chunk: {chunk}

Respond in the following format:
Line 1: "YES" or "NO" - indicate whether the chunk is relevant to the query.
Line 2: Your analysis reasoning, briefly explaining why it is relevant or irrelevant.

Please strictly adhere to this 2-line format with no additional text, explanations, or commentary.
"""
```

Figure 5: Automatic Annotation Prompt Template (Details partially omitted due to space constraints).

D.2 Annotation accuracy

To ensure the high quality and reliability of our in-house company datasets (Zeekr and Lotus), we adopted a rigorous annotation process based on our automatic annotation prompt template. In the first stage, we conducted a pilot annotation on a randomly selected subset of samples from both the Lotus and Zeekr datasets. The automatically generated annotations for these subsets were then subjected to manual review by domain experts. This manual quality check yielded high accuracy scores, confirming the initial effectiveness of the prompt. For challenging edge-case samples identified during this review, we further enhanced the prompt by integrating them as new few-shot examples. In the second stage, we leveraged this refined prompt template to annotate the entirety of the two datasets, ensuring consistent and high-quality relevance labels across the full data distribution. The quality assessment results are detailed in Table 3.

E Latency and Cost Analysis

We conduct a detailed analysis of our system's operational efficiency, focusing on two critical aspects: **system latency** and **usage**

Table 3: Annotation Accuracy on Dataset Subsets.

Dataset	Subset Size (chunks)	Accuracy (%)
Lotus	1350	95.5
Zeekr	1296	92.8

cost. We first measure the average execution time for each process within our system. Crucially, we also quantify the average token usage for each call to the Chat Completion API, which serves as the primary factor in estimating the marginal deployment cost of the system. A deeper analysis is necessary for the asynchronous and variable components, particularly **Sub-query Answering** and **Tool Use**, since the system performance is dependent on the number of sub-queries executed and the number of available external tools. The system performance is detailed in Table 4.

Table 4: System Latency and Deployment Cost. t and n ($n \geq 1$) denote the number of available tools and the number of sub-queries, respectively.

Step	Process	Time (s)	Model Used	Token Usage	Cost Estimation (per k queries)
1	Query rewriting	~ 2.4	DeepSeek-V3 API	~ 600	~ \$0.17
2	Retrieval & re-ranking	~ 1.8	Locally Deployed	N/A	24GB GPU
	Tool Use (async, opt.)	Varies	DeepSeek-V3 API	~ $250 \times n + 100 \times t \times n$	~ $\$0.07 \times n + \$0.02 \times t \times n$
3	Sub-query answering (async)	~ 4.7	DeepSeek-V3 API	~ $4500 \times n$	~ $\$1.3 \times n$
4	Final answer merging	~ 1.6	DeepSeek-V3 API	~ $200 \times (n - 1)$	~ $\$0.14 \times (n - 1)$
Total	—	~ 10.5	—	~ $400 + 4.95k \times n + 100 \times n \times t$	~ $\$0.11 + \$1.4 \times n + \$0.028 \times n \times t$

F RAGAS

To comprehensively evaluate retrieval and generation performance, we employ the RAGAS framework [14], which provides automated, reference-based metrics for RAG systems.

Factual Correctness. metric assesses the factual overlap between a generated response and a reference (ground-truth) answer. Internally, it uses an LLM as the evaluator to decompose both the response and the reference into sets of atomic claims. It then employs natural language inference to judge which response claims correspond to reference claims. A higher score indicates that the generated response preserves more ground-truth facts.

Answer Relevancy. Answer Relevancy quantifies how pertinent or directly relevant the generated answer is to the user’s query. RAGAS implements it to penalize responses that are incomplete, extraneous, or tangential, awarding a higher score to answers that closely align in substance and focus with the original question.

Context Recall. Context Recall measures the extent to which the retrieved context covers all the claims needed to answer the question. In RAGAS, the reference answer is decomposed into claims, and the metric counts how many of those claims are supported by the retrieved context, using LLM-based attribution when necessary. Higher recall indicates fewer missing supports.

Context Precision. This metric evaluates how well the retriever ranks relevant information above irrelevant passages. Specifically, it measures whether the ground-truth relevant items present in the contexts are ranked higher in the retrieve list, using a mean

precision@k scheme over retrieved chunks. A high CP means that relevant chunks tend to appear at top ranks rather than buried.

G LightRAG and GraphRAG

G.1 GraphRAG Configuration

GraphRAG is a graph-based retrieval system that uses knowledge graph representations to enhance contextual understanding. The configuration is detailed in Table 5.

LLM	
Parameter	Value
Model	deepseek-v3
Message Caching	Enabled(hash-based KV)
Embedding	
Model	all-MiniLM-L6-v2
Dimension	384
Max Token Size	8192
Batch Size	64
Max Async	4
Normalize	True
Retrieval / Query	
Mode	local
Top cited chunks	10 (EVAL_TOP_CHUNKS)
Storage	
Vector DB	NanoVectorDB
Graph Storage	NetworkX + GraphML
KV Storage	JSON

Table 5: Graphrag configuration

G.2 LightRAG Configuration

LightRAG represents a more streamlined approach which prioritizes efficiency while maintaining retrieval quality. The detailed configuration is shown in Table 6.

LLM	
Parameter	Value
Model	deepseek-v3
Configuration	Custom LLM model func
Embedding	
Model	all-MiniLM-L6-v2
Dimension	384
Max Token Size	8192
Embedding batch num	50
Max parallel insert	2
Query	
Mode	global
Top_k	60
Response type	Multiple Paragraphs
Only need context	False
Max token for text unit	4000
Max token for global context	4000
Max token for local context	4000

Table 6: LightRAG Configuration